Q1.are the classes equally represented?

yes, the classes are almost equally represented.

class 0 contains 35,021 rows (50.03%), while class 1 contains 34,979 rows (49.97%).

the difference is only 42 rows, indicating a balanced target distribution with no significant class imbalance.

Q2. where are values concentrated, how wide is the range, and are there skewness, gaps, spikes, or multiple peaks?

- concentration:
  - systolic blood pressure (`ap_hi`): most values are concentrated at 120.
  - diastolic blood pressure (`ap_lo`): most values are concentrated at 80.
  - pulse pressure (`pulse_pressure`): most values are concentrated at 40.
  - mean arterial pressure (`map`): many values are concentrated around 93.33.
- range and skewness:
  - age (`age_years`): ranges from 30 to 65 years, with a mean of 53.3 and a median of 54.
  - height (`height`): ranges from 120 to 207 cm, with a mean of 164.4 cm.
  - weight (`weight`): ranges from 30 to 200 kg, with a mean of 74.2 kg.
  - body mass index (`bmi`): ranges from 10.73 to 108.17 and is right-skewed because of some very high values.

Q3. which categories are common or rare, and is an unknown category substantial?

- **gender:** category 1 has more rows (45,530) than category 2 (24,470).
- **cholesterol:** most people are in category 1 (52,385 rows), while category 3 has the lowest count (8,066 rows).
- **gluc:** category 1 is the most common (59,479 rows), while categories 2 and 3 have much lower counts.
- **smoke:** most people are non-smokers (63,831 rows), compared to 6,169 smokers.
- **alco:** most people do not consume alcohol (66,236 rows), and only 3,764 consume alcohol.
- **active:** most people are physically active (56,261 rows), while 13,739 are inactive.

overall, some categories are much more common than others, especially in glucose, smoking, and alcohol consumption. no separate unknown category appears in these plots.

Q4.which measurements have unusual values relative to their distribution?

- **age_years:** Features an unusual value on the lower end (around 30 years old).
- **height:** Features multiple flagged values on both the lower end (approx. 120–135) and the upper end (approx. 185–207).
- **weight:** Features numerous flagged values on the lower end (<50) and an extended tail of high values ranging from approximately 110 to 200.
- **ap_hi:** Features flagged values on both the lower end (~60–90) and the higher end (~175–240).
- **ap_lo:** Features flagged values on both the lower end (<60) and the higher end (~100–180).

Q5. does each class occupy a different part of a measurement's range?

- **height:** both classes have almost the same distribution, with a lot of overlap between them.
- **age_years:** both classes range from around 38 to 65 years. class 1 has more older patients, while class 0 is more concentrated at younger ages, but both classes overlap.
- **weight:** both classes have a wide range of values. class 1 tends to have higher weights, but there is still a lot of overlap.
- **ap_hi:** class 1 tends to have higher systolic blood pressure values than class 0, but both classes share similar ranges.
- **ap_lo:** class 1 also tends to have higher diastolic blood pressure values, although the distributions overlap considerably.

overall, age and blood pressure show some differences between the classes, while height has very similar distributions. no single measurement can clearly separate the two classes because their values overlap.

Q6. how do the average and observed extremes differ between target categories?

- **age_years:** class 1 has a higher average age (around 55 years) than class 0 (around 52 years). class 0 has a lower minimum age, but both classes have almost the same maximum age.
- **height:** both classes have almost the same mean, minimum, and maximum height, so there is no big difference between them.
- **weight:** class 1 has a higher average weight (around 76 kg) than class 0 (around 71 kg), but both classes have similar minimum and maximum values.
- **ap_hi:** class 1 has a higher average systolic blood pressure (around 134) than class 0 (around 119). class 0 has a lower minimum, but both classes have a similar maximum.
- **ap_lo:** class 1 has a higher average diastolic blood pressure (around 84) than class 0 (around 78). both classes have a similar minimum, but class 1 has a higher maximum.

overall, class 1 has higher average age, weight, and blood pressure than class 0, while height is almost the same in both classes. these differences show a relationship with cardiovascular disease, but they do not mean that these features are the direct cause.

Q7. how many rows from each class occur in each category?

- **age_years:** cardio = 1 has a higher average age (around 55 years) than cardio = 0 (around 52 years). cardio = 0 has a lower minimum age, but both have almost the same maximum age.
- **height:** both classes have almost the same average, minimum, and maximum height, so there is no big difference between them.
- **weight:** cardio = 1 has a higher average weight (around 76 kg) than cardio = 0 (around 71 kg). both classes have the same minimum and maximum values.
- **ap_hi:** cardio = 1 has a higher average systolic blood pressure (around 134) than cardio = 0 (around 119). cardio = 0 has a lower minimum, but both have similar maximum values.
- **ap_lo:** cardio = 1 has a higher average diastolic blood pressure (around 84) than cardio = 0 (around 78). both classes have the same minimum, but cardio = 1 has a higher maximum.

overall, cardio = 1 has higher average age, weight, and blood pressure, while height is almost the same in both classes. these differences show a relationship with cardiovascular disease, but they do not prove that these features cause it.